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EMT 325 — Past Questions & Materials
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Course: EMT 325
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 300

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Paper 1 of 4 — EMT 325 Introductory Toxicology 2024

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE , |

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT & TOXICOLOGY

2023/2024 ACADEMIC SESSION 2%" SEMESTER EXAMS. oS

EMT 325: INTRODUCTORY TOXICOLOGY 2 UNITS

INSTRUCTION: ATTEMPT ANY FOUR QUESTIONS TIME: 2 HOURS

Question 1

(a) Define toxicology.

(b) Extensively discuss the historical development of toxicology.

Question 2

a) Write a short note on the Toxicity Characteristics Leaching Procedure (TCLP).

b) i) List three physical, three chemical, and three biological methods for managing toxic waste in the environment.

ii) Explain one of the physical, chemical, or biological methods listed above.

Question 3

a) List five factors to consider when selecting a pesticide.

b) Describe how to protect your face and eyes from exposure during pesticide application. ,

c) Write a short note on the impact of pesticide use on human health.

« Question 4

Discuss a global disaster related to the m

headings:

i. Nature of the misuse or abuse.

ij. Environmental impacts resulting from t

Health impacts caused by the misuse or abuse.

misuse and abuse of pesticides under the following

he misuse or abuse.

iii.

Question 5

Discuss the bioc

the soil matrix.

hemical properties of pesticides that affect their mobility and transport within

Question 6

and abuse.

Suggest at least six strategies to address the challenges of pesticide misuse

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Paper 2 of 4 — EMT 325 pesticides enzymatic and non enzymatic conversion

Level 300 • Second Semester • Handout

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Enzymatic and Non-Enzymatic Conversion of Pesticides in the Environment

Enzymatic Conversion

1. **Microbial Enzymes:** Microorganisms such as bacteria, fungi, and algae play a crucial role in the enzymatic degradation of pesticides. These organisms produce enzymes that catalyze the breakdown of pesticide molecules into less harmful substances.
 - **Dehalogenases:** These enzymes remove halogen atoms from organochlorine pesticides, making them less toxic.
 - **Phenol Hydroxylase and Laccases:** These enzymes oxidize phenolic compounds, aiding in the degradation of various pesticides.
 - **Oxidoreductases:** Enzymes like cytochrome P450 monooxygenases introduce oxygen into pesticide molecules, increasing their solubility and facilitating further breakdown.
2. **Bioremediation Phases:**
 - **Phase I:** Involves oxidation-reduction and hydrolysis reactions that increase the solubility of pesticide compounds.
 - **Phase II:** Involves conjugation reactions that transform toxic pollutants into less toxic or non-toxic products.

Non-Enzymatic Conversion

1. **Photodegradation:** Sunlight, particularly UV radiation, can break down pesticide molecules. This process, known as photolysis, involves the absorption of light energy by pesticide molecules, leading to their decomposition.
2. **Chemical Hydrolysis:** Pesticides can undergo hydrolysis, a reaction with water, resulting in the breakdown of the pesticide into less harmful substances. This process is influenced by factors such as pH and temperature.
3. **Oxidation and Reduction:** Chemical reactions involving the transfer of electrons can alter the structure of pesticide molecules. Oxidation typically involves the addition of oxygen or the removal of hydrogen, while reduction involves the gain of electrons.
4. **Adsorption and Desorption:** Pesticides can adhere to soil particles (adsorption) or be released from them (desorption). These processes affect the mobility and bioavailability of pesticides in the environment.

Conclusion

Both enzymatic and non-enzymatic processes are essential for the conversion and degradation of pesticides in the environment. Enzymatic processes, driven by microbial activity, offer a biological approach to detoxifying pesticides, while non-enzymatic processes rely on physical and chemical reactions to achieve similar outcomes. Together, these mechanisms help mitigate the environmental impact of pesticide use.

Paper 3 of 4 — Pesticide Hoe EMT 325

Level 300 • Second Semester • Handout

Pesticide Hoe;

Choosing the right pesticide hoe (application equipment) depends on several factors. Consider the following:

1. Type of pesticide: Different equipment is designed for specific pesticide formulations (e.g., liquids, granules, powders).
2. Target area: Select equipment suitable for the size and shape of the area to be treated (e.g., small plots, large fields, orchards).
3. Crop or plant type: Choose equipment designed for the specific crop or plant being treated (e.g., row crops, trees, vines).
4. Application method: Consider the desired application method (e.g., spraying, dusting, fogging).
5. Operator safety: Select equipment with features that minimize exposure risk (e.g., enclosed cabs, protective shields).
6. Environmental concerns: Consider equipment that reduces drift, runoff, or other environmental impacts.
7. Cost and maintenance: Balance equipment cost with maintenance needs and durability.
8. Regulatory requirements: Ensure the equipment meets local regulations and guidelines.

ENVIRONMENTAL CONSIDERATIONS IN CHOOSING A PESTICIDE HOE

When choosing a pesticide hoe (application equipment), consider the following environmental concerns:

1. Drift reduction: Look for equipment with features that minimize drift, such as drift-reducing nozzles or shields.
2. Spray droplet size: Choose equipment that produces larger droplet sizes to reduce drift and improve targeting.
3. Buffer zones: Ensure equipment can maintain buffer zones around sensitive areas, like water bodies or wildlife habitats.
4. Soil contamination: Consider equipment that reduces soil contamination, such as precision applicators or injectors.
5. Water protection: Choose equipment that prevents water contamination, like closed systems or containment vessels.
6. Air pollution: Opt for equipment with low emissions or alternative propulsion methods, like electric or battery-powered devices.
7. Waste reduction: Select equipment that minimizes waste, such as reusable containers or efficient application systems.
8. Environmental monitoring: Consider equipment with built-in monitoring systems to track environmental impact.

9. Integrated Pest Management (IPM): Choose equipment that supports IPM practices, like precision application and monitoring tools.

10. Sustainability: Prioritize equipment with sustainable design, energy efficiency, and recyclable materials.

TYPES OF PESTICIDE HOE

Sprayer is a machine used to apply liquid chemicals on plants to control pest and diseases. It can also be used to apply herbicides to control weeds and to spray micro-nutrients to enhance plant growth.

The main functions of a sprayer are

- Breaking the chemical solution in to fine droplets of effective size.
- Distributing the droplets uniformly over the plants.
- Applying the chemicals with sufficient pressure for positive reaching the plants
- Regulating the amount of liquid applied on plants to avoid excessive application.

A good sprayer should posses the following qualities

- It should produce a steady stream of spray material in desired droplet size so that the plants to be treated may be covered uniformly.
- It should deliver the liquid at sufficient pressure so that the spray solution reaches all the foliage and spreads uniformly over the plant body.
- It should be light in weight yet sufficiently strong, easily workable and repairable.

SPRAYERS

prayers are a popular choice for pesticide application! Here are some types of sprayers:

1. Backpack Sprayers:

- Portable, wearable tanks (usually 2-4 L)
- Ideal for small to medium-sized areas, gardens, and landscaping

2. Handheld Sprayers:

- Small, portable sprayers (typically 1-2 L)
- Suitable for small areas, spot treatments, and precision applications

3. Tractor-Mounted Sprayers:

- Large, tank-mounted sprayers (often 10-100 gallons)
- Designed for extensive agricultural or commercial applications

4. Aerial Sprayers:

- Plane-mounted or helicopter-mounted sprayers
- Used for large-scale agricultural or forestry applications

6. * Orchard Sprayers*:

- Specialized sprayers for orchard applications
- Often feature telescoping booms and precision nozzles

DUSTERS

Duster is a machine used to apply chemicals in dust form. Dusters make use of air stream to carry pesticides in finely divided form on the plants. Dusters are designed to distribute the pesticide evenly and efficiently, ensuring effective coverage and minimizing waste.

Types of dusters include:

1. Plunger type - it is a simple duster with a small piston. The piston drives a current of air over the dust in the hopper. The dust is carried away through a delivery spout. Small hand pump dusters of this type are available and are suitable only where the area to be dusted is small like vegetable gardens
2. Knapsack type - It is a duster with the powder container carried on the back of the operator. Knapsack dusters have a hopper through which a current of air is blown to pick up the dust. The air current is produced by a lever operated leather bellows. Shoulder straps are used to carry in the field. These dusters are suitable for small areas.
3. Rotary duster - Hand rotary dusters are useful to apply chemicals which are in powder form. It consists of a hopper, a fan, gear box, handle, delivery hose and a deflector plate.

When the handle is rotated the fan rotates at high speed and draws air from outside. The chemical from hopper is fed in to the air stream in the suction side of the fan. The chemical mixes with the air, passes through the delivery line and is applied on the plants. The rate of delivery can be regulated. It is used to apply powdery chemicals to vegetables, sorghum etc. crops.

Aerial duster or crop duster - an aircraft is used for dusting or spraying large acreages with pesticides. Aerial spraying and dusting permit prompt coverage of large areas at the moment when application of pesticide is most effective and avoid the need for wheeled vehicles that might damage crops.

FOGGERS

Foggers, in the context of pesticidal uses, refer to equipment or devices used to apply pesticides or other substances in a fog or mist form. Foggers are designed to create a fine spray that can penetrate dense foliage, reach hidden areas, and provide effective coverage.

Thermal Foggers

A thermal fogger is a device you can use to fight pest problems in outdoor areas. As the name suggests, a thermal fogger uses heat to vaporize a fogging solution. In most cases, this liquid is an insecticide. It sprays the solution out of the nozzle of the fogger in the form of a mist or a fog. Thermal fogging is a highly specialised form of pesticide application. Liquid pesticide formulation subjected to ultrahigh temperature in the fogging machine is converted into a vapour and released through a fogging pipe into the atmosphere. On interfacing with cold air, the vapour condenses into liquid to form clouds of tiny droplets which collectively form a fog.

Be that as it may, hot (thermal) fogging has been successfully used in the field on tropical tree and bush crops in estates and plantations for many decades by taking advantage of special features of canopy structure and atmospheric conditions at certain times of the day. The fog is kept within the canopy and close to the ground for a sufficiently long period to allow droplets to deposit on the foliage for insect pest and disease control as appropriate.

Cocoa and rubber Cocoa and rubber are two tropical tree crops in which special features of canopy form and structure have enabled thermal fogging to be used for insect pest and disease control.

Cold Foggers or Ultra low Volume (ULV) foggers

Cold Foggers or Ultra low Volume (ULV) foggers provide a cold fog with particle sizes in the 5-20 micron range. Cold foggers use air pressure to atomize the insecticide and are powered by gasoline, compressed air, or electricity. The lack of heat (and thus the fire risk) makes the ULV fogger safer for indoor use.

Both thermal foggers and ULV or cold foggers may be used inside or outside and both can fog water or oil-based products. Sanitizers, disinfectants, mold, and fungi control products are most often applied by cold foggers which allow adjustment of the particle size and provide larger particles used for these applications.

Research data indicate ULV fog is best for indoor control of stored product insects (beetles, moths, etc.). The optimum particle size theory plays an important role in the use of ULV fogging equipment for insect control. Droplets of pesticide found to be most effective are those with a small size range allowing them to remain airborne for a considerable length of time and to be carried to the insect by atmospheric transport

Injectors

Injectors are specialized equipment used to apply substances, such as pesticides, fertilizers, or soil amendments, directly into the soil or plant tissues. They are designed to deliver the material precisely and efficiently, minimizing waste and environmental impact.

Types of injectors:

1. Soil injectors: Apply substances directly into the soil, often used for fertilizers and soil amendments.
2. Foliar injectors: Apply substances directly onto plant leaves, often used for pesticides and fertilizers.
3. Tree injectors: Apply substances directly into tree trunks or roots, often used for pest control and fertilization.
4. Precision injectors: Use advanced technologies, such as GPS and sensors, to optimize application rates and placement.

Precision Applicators

Precision applicators are advanced tools designed to apply substances, such as pesticides, fertilizers, or seeds, with high accuracy and precision. They use various technologies to optimize application rates, placement, and timing, minimizing waste and environmental impact.

Types of precision applicators:

1. Precision sprayers: Use advanced nozzle technology and sensors to optimize spray patterns and droplet size.
2. Drones with spraying capabilities: Utilize GPS and sensors to apply substances with precision.
3. Autonomous farming equipment: Use AI, GPS, and sensors to optimize application rates and placement.
4. Variable rate applicators: Adjust application rates based on soil type, moisture levels, and crop density.
5. Smart spreaders: Use sensors and GPS to optimize spreading patterns and rates

Paper 4 of 4 — EMT 325 lecture Notes

Level 300 • Second Semester • Handout

MANAGEMENT OF HAZADOUS WASTE

1. Introduction

Waste was associated with human society from prehistory to today and no doubt will continue for the future. People have to manage the produced waste. Disposal of waste into the surrounding locality has to date been the usual practice with little concern for the environment. Waste has to be managed properly to preserve the planet for the coming generations. Waste generally generated accordingly with life continuity and related proportionally with the human activities such as agricultural, industrial, residential, institutional, municipal, commercial, mining, recreational, and others. This issue is strongly increasing and becomes a potential trouble in the community. The main focus of this study is on hazardous and radioactive wastes accompanying with their Different technologies developed for management. Rapid trend of industry and high-technological progress are the main sources of the accumulation of hazardous materials. Nuclear applications have been rapidly developed recently, and several nuclear power plants have been started to work throughout the world. The potential impact of released radioactive contaminants into the environment has received growing attention due to nuclear accidents, which pose serious problems to biological systems.

Hazardous wastes are those that may contain toxic substances generated from industrial, hospital, some types of household wastes. These wastes could be corrosive, inflammable, explosive, or react when exposed to other materials. Some hazardous wastes are highly toxic to environment including humans, animals, and plants. The management of extremely increasing volumes of these wastes became a very important accordingly. Inadequate management of waste led to contamination of environment: water, soil, and atmosphere and to a serious impact on public health. Direct health impacts of mismanagement of waste are well known and can be observed obviously in developing countries. Saving of the environment and human health from the detrimental effects of hazardous and radioactive wastes is achieved by the effective improvement of waste management programs.

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2. Hazardous wastes

With increasing manufacturing processes, solid, liquid, and/or gaseous emissions generate as by-products. Some of these wastes are potentially harmful to human health and environment and thus need special techniques of management.

Wastes are classified as hazardous if they exhibit one or more of ignitability, corrosivity, reactivity, or toxicity. According to Resource Conservation and Recovery Act (RCRA), hazardous wastes are defined as any waste or combination of wastes which pose a substantial present or potential hazard to human health or living organisms because

such wastes are non-degradable or persistent in nature or because they can be biologically magnified, or because they can be lethal, or because they may otherwise cause or tend to cause detrimental cumulative effects.

The management of hazardous wastes has become a specialized discipline because of the complex nature of the problem and the solutions available to humanity. The mismanagement examples of hazardous wastes causing disastrous human and environmental consequences are numerous. The management process is based on the definition and classification of the different wastes, and their toxic effects on human and taking in consideration the application of risk management to control human health and environmental impacts of hazardous waste. Hazardous waste management, therefore, deals with minimizing harmful effects on humans and environment by applying special techniques of handling, storage, transportation, treatment, and disposal of hazardous wastes.

3. Hazardous characteristics

A useful listing of hazardous characteristics is that provided by the United Nations [2] as part of recommendations relating to the transport of dangerous goods as illustrated in Table 1.

UN class Hazardous characteristic
number

2

1 Explosive

3, 4 Flammable

5 Oxidizing

6 Poisonous/infectious

7 Radioactive

8 Corrosive

9 Toxic (delayed or
chronic)/ecotoxic

Table 1.

Hazardous characteristics: extracted from UN listing [2].

3.1. Industrial wastes

Waste generated from industrial sources can have non-hazardous and hazardous components, with non-hazardous waste usually representing the greater part of the volume. The hazardous component of this waste is relatively small in volume [3].

This type of waste was identified as hazardous waste when proceeds toxicity test, corrosively test, ignitability test, and some special character test. As a hazardous pollutant, it may impose serious impacts on surrounding environment and such impacts should be quantitatively examined to assess the influence on human health [4].

3.2. Household waste

Households generate small quantities of hazardous wastes such as oil-based paints, paint thinners, wood preservatives, pesticides, insecticides, household cleaners, used motor oil, antifreeze, and batteries. It has been estimated that household hazardous waste in industrialized countries such as the United States accounts for a total of about 0.5% (by weight) of all waste generated at home, while in most developing countries, the percentage probably is even lower [3].

3.3. Biomedical waste [5]

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There are some of hazardous medical and dental wastes that, when disposed improperly, could cause harm to the environment. It also presents an occupational health hazards to the healthcare personnel who handle these wastes at the point of generation and those involved with their management, that is, segregation, storage, transport, treatment, and disposal.

Healthcare waste that is capable of producing injury or disease including many sorts of hazardous wastes such as:

- Infectious waste: Which contain pathogens namely bacteria, viruses, fungi, or parasites in concentrations sufficient to cause disease in susceptible hosts. Cultures and stocks of infectious agents from laboratory work; tissues and dressing generated from autopsies, surgeries, and treatment of infected patients and animals; materials or equipment in contact with blood and infected body fluids.
- Pathological waste: Including tissue, organs, body parts, human fetuses, and animal carcasses, blood and body fluids.
- Sharps: It comprise syringes, needles, scalpels, saws, infusion sets, knives, blades, broken glass, or other items that can cause cut or puncture wounds.
- Pharmaceutical waste: It covers expired, unused, spilt, and contaminated pharmaceutical products, drugs, vaccines, and sera that are no longer required and need to be disposed of in appropriate manner.
- Genotoxic waste: This type combines cytostatic drugs, vomit, urine, or feces from the patients treated with cytotoxic drugs, chemicals, and radioactive materials. Genotoxic waste has mutagenic, teratogenic, and carcinogenic properties.
- Chemical waste: Discarded solid, liquid, or gaseous chemicals should be considered as hazardous if it is toxic, corrosive, inflammable, or reactive.
- Waste with high content of heavy metals: Mercury (thermometers, blood pressure gauges, amalgam), cadmium (discarded batteries), and lead (reinforced wood panels for radiation proofing in radiology department) generated from hospitals could be represented as a subcategory of hazardous chemical waste.

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- Radioactive waste: The use of radioisotopes in vitro analysis of body tissues and fluids, in vivo organ imaging, tumor localization, and treatment and various

clinical studies involving certain radionuclides need to be specially managed in a centralized treatment facility for radioactive wastes.

3.4. Radioactive waste

Nuclear applications have been rapidly developed recently, and several nuclear power plants started to work throughout the world. The potential impact of released radioactive contaminants into the environment has received growing attention due to nuclear accidents. Contamination of soil and water by radionuclides due to natural processes, global fallout from nuclear weapon testing, discharges from nuclear installations, disposal of nuclear waste, and occasional nuclear accidents poses serious problems to biological systems. Radioactive waste includes a variety of radionuclides and occurs in a variety of physical and chemical forms. It can be generally classified as low-/intermediate-level radioactive waste and high-level radioactive waste. Radioactive waste, arising from civilian nuclear activities as well as from weapon activities, poses a potential problem for handling and saving the environment for coming generations.

Radioactive waste includes a variety of radionuclides and occurs in a variety of physical and chemical forms. It can be generally classified as low-/intermediate-level radioactive waste and high-level radioactive waste. Nuclear research establishments include, for example, waste containing different organic components, toxic or chemically aggressive constituents, radionuclides with specific properties (high mobility, high chemical activity, volatile elements, etc.), waste difficult for treatment and not appropriate for direct immobilization (e.g., spent organic ion exchange resins and spent liquid scintillation cocktails). For such waste, application of conventional treatment and conditioning options may not be efficient and appropriate in terms of economy, safety, and performance characteristics. In many cases, such wastes are stored awaiting an appropriate treatment and conditioning solution. The primary sources of radioactive wastes in a country without nuclear fuel cycle activities are nuclear research, production

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of radioisotopes, application of radioisotopes, and decontamination and decommission of nuclear installations.

Treatment, storage, and disposal OF HAZARDOUS WASTE

Several options are available for hazardous-waste management. The most desirable is to reduce the quantity of waste at its source or to recycle the materials for some other productive use. Nevertheless, while reduction and recycling are desirable options, they are not regarded as the final remedy to the problem of hazardous-waste disposal. There will always be a need for treatment and for storage or disposal of some amount of hazardous waste.

Treatment

Hazardous waste can be treated by chemical, thermal, biological, and physical methods. Chemical methods include ion exchange, precipitation, oxidation and reduction, and neutralization. Among thermal methods is high-temperature incineration,

which not only can detoxify certain organic wastes but also can destroy them. Special types of thermal equipment are used for burning waste in either solid, liquid, or sludge form. These include the fluidized-bed incinerator, multiple-hearth furnace, rotary kiln, and liquid-injection incinerator. One problem posed by hazardous-waste incineration is the potential for air pollution.

Biological treatment of certain organic wastes, such as those from the petroleum industry, is also an option. One method used to treat hazardous waste biologically is called landfarming. In this technique the waste is carefully mixed with surface soil on a suitable tract of land. Microbes that can metabolize the waste may be added, along with nutrients. In some cases a genetically engineered species of bacteria is used. Food or forage crops are not grown on the same site. Microbes can also be used for stabilizing hazardous wastes on previously contaminated sites; in that case the process is called bioremediation.

The chemical, thermal, and biological treatment methods outlined above change the molecular form of the waste material. Physical treatment, on the other hand,

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concentrates, solidifies, or reduces the volume of the waste. Physical processes include evaporation, sedimentation, flotation, and filtration. Yet another process is solidification, which is achieved by encapsulating the waste in concrete, asphalt, or plastic. Encapsulation produces a solid mass of material that is resistant to leaching. Waste can also be mixed with lime, fly ash, and water to form a solid, cementlike product.

Surface storage and land disposal

Hazardous wastes that are not destroyed by incineration or other chemical processes need to be disposed of properly. For most such wastes, land disposal is the ultimate destination, although it is not an attractive practice, because of the inherent environmental risks involved. Two basic methods of land disposal include landfilling and underground injection. Prior to land disposal, surface storage or containment systems are often employed as a temporary method.

Temporary on-site waste storage facilities include open waste piles and ponds or lagoons. New waste piles must be carefully constructed over an impervious base and must comply with regulatory requirements similar to those for landfills. The piles must be protected from wind dispersion or erosion. If leachate is generated, monitoring and control systems must be provided. Only non-containerized solid, nonflowing waste material can be stored in a new waste pile, and the material must be landfilled when the size of the pile becomes unmanageable.

A common type of temporary storage impoundment for hazardous liquid waste is an open pit or holding pond, called a lagoon. New lagoons must be lined with impervious clay soils and flexible membrane liners in order to protect groundwater. Leachate collection systems must be installed between the liners, and groundwater monitoring

wells are required. Except for some sedimentation, evaporation of volatile organics, and possibly some surface aeration, open lagoons provide no treatment of the waste. Accumulated sludge must be removed periodically and subjected to further handling as a hazardous waste.

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Many older, unlined waste piles and lagoons are located above aquifers used for public water supply, thus posing significant risks to public health and environmental quality. A large number of these old sites have been identified and scheduled for cleanup, or remediation, around the world.

Secure landfills

Landfilling of hazardous solid or containerized waste is regulated more stringently than landfilling of municipal solid waste. Hazardous wastes must be deposited in so-called secure landfills, which provide at least 3 metres (10 feet) of separation between the bottom of the landfill and the underlying bedrock or groundwater table. A secure hazardous-waste landfill must have two impermeable liners and leachate collection systems. The double leachate collection system consists of a network of perforated pipes placed above each liner. The upper system prevents the accumulation of leachate trapped in the fill, and the lower serves as a backup. Collected leachate is pumped to a treatment plant. In order to reduce the amount of leachate in the fill and minimize the potential for environmental damage, an impermeable cap or cover is placed over a finished landfill.

A groundwater monitoring system that includes a series of deep wells drilled in and around the site is also required. The wells allow a routine program of sampling and testing to detect any leaks or groundwater contamination. If a leak does occur, the wells can be pumped to intercept the polluted water and bring it to the surface for treatment.

One option for the disposal of liquid hazardous waste is deep-well injection, a procedure that involves pumping liquid waste through a steel casing into a porous layer of limestone or sandstone. High pressures are applied to force the liquid into the pores and fissures of the rock, where it is to be permanently stored. The injection zone must lie below a layer of impervious rock or clay, and it may extend more than 0.8 km (0.5 mile) below the surface. Deep-well injection is relatively inexpensive and requires little or no pretreatment of the waste, but it poses a danger of leaking hazardous waste and eventually polluting subsurface water supplies.

Remedial action

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Disposal of hazardous waste in unlined pits, ponds, or lagoons poses a threat to human health and environmental quality. Many such uncontrolled disposal sites were used in the past and have been abandoned. Depending on a determination of the level of risk, it may be necessary to remediate those sites. In some cases, the risk may require emergency action. In other instances, engineering studies may be required to assess

the situation thoroughly before remedial action is undertaken.

One option for remediation is to completely remove all the waste material from the site and transport it to another location for treatment and proper disposal. This so-called off-site solution is usually the most expensive option. An alternative is on-site remediation, which reduces the production of leachate and lessens the chance of groundwater contamination. On-site remediation may include temporary removal of the hazardous waste, construction of a secure landfill on the same site, and proper replacement of the waste. It may also include treatment of any contaminated soil or groundwater. Treated soil may be replaced on-site and treated groundwater returned to the aquifer by deep-well injection.

A less costly alternative is full containment of the waste. This is done by placing an impermeable cover over the hazardous-waste site and by blocking the lateral flow of groundwater with subsurface cutoff walls. It is possible to use cutoff walls for this purpose when there is a natural layer of impervious soil or rock below the site. The walls are constructed around the perimeter of the site, deep enough to penetrate to the impervious layer. They can be excavated as trenches around the site without moving or disturbing the waste material. The trenches are filled with a bentonite clay slurry to prevent their collapse during construction, and they are backfilled with a mixture of soil and cement that solidifies to form an impermeable barrier. Cutoff walls thus serve as vertical barriers to the flow of water, and the impervious layer serves as a barrier at the bottom.

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PRINCIPLES OF PEST CONTROL

A pest is anything that:

- competes with humans, domestic animals, or desirable plants for food or water,
- injures humans, animals, desirable plants, structures, or possessions,
- spreads disease to humans, domestic animals, wildlife, or desirable plants,
- annoys humans or domestic animals.

Types of Pests

Types of pests include:

- insects, such as roaches, termites, mosquitoes, aphids, beetles, fleas, and caterpillars.
- insect-like organisms, such as mites, ticks, and spiders,
- microbial organisms, such as bacteria, fungi, nematodes, viruses, and mycoplasmas,
- weeds, which are any plants growing where they are not wanted,
- mollusks, such as snails, slugs, and shipworms, and
- vertebrates, such as rats, mice, other rodents, birds, fish, and snakes.

Most organisms are not pests. A species may be a pest in some situations and not in others. An organism should not be considered a pest until it is proven to be one.

Categories of pests include:

- continuous pests that are nearly always present and require regular control.
- sporadic, migratory, or cyclical pests that require control occasionally or intermittently.
- potential pests that do not require control under normal conditions, but may require control in certain circumstances.

Pest Identification

Accurate identification is the first step in an effective pest management program.

Never attempt a pest control program until you are sure of what the pest is. The more you know about the pest and the factors that influence its development and spread, the easier, more cost-effective, and more successful your pest control will be. Correct

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identification of a pest allows you to determine basic information about it, including its life cycle and the time that it is most susceptible to being controlled.

As a certified applicator, you must be familiar with the pests you are likely to encounter in the type of work in your certification category. To be able to identify and control pests, you need to know:

- the physical features of the pests likely to be encountered.
 - characteristics of the damage they cause,
 - their development and biology,
 - whether they are continuous, sporadic, or potential pests, and
 - what your control goal is.
- If you need help in identifying a pest, contact your commodity or industry organization, Cooperative Extension agent, or State land grant university.

Pest Control

Any time you are considering whether pest control is necessary, remember:

- Control a pest only when it is causing or is expected to cause more harm than is reasonable to accept.
- Use a control strategy that will reduce the pest numbers to an acceptable level.
- Cause as little harm as possible to everything except the pest.
- Even though a pest is present, it may not do very much harm. It could cost more to control the pest than would be lost because of the pest's damage.

Pest Control Goals

Whenever you try to control a pest you will want to achieve one of these three goals, or some combination of them:

- prevention - keeping a pest from becoming a problem.
- suppression - reducing pest numbers or damage to an acceptable level, and .
- eradication - destroying an entire pest population.

a. Prevention may be a goal when the pest's presence or abundance can be predicted in advance. Continuous pests, by definition, are usually very

predictable. Sporadic and potential pests may be predictable if you know the circumstances or conditions that will favor their presence as pests. For example,
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some plant diseases occur only under certain environmental conditions. If such conditions are present, you can take steps to prevent the plant disease organisms from harming the desirable plants.

b. Suppression is a common goal in many pest situations. The intent is to reduce the number of pests to a level where the harm they are causing is acceptable. Once a pest's presence is detected and the decision is made that control is necessary, suppression and prevention often are joint goals. The right combination of control measures can often suppress the pests already present and prevent them from building up again to a level where they are causing unacceptable harm.

c. Eradication is a rare goal in outdoor pest situations, because it is difficult to achieve. Usually the goal is prevention and/or suppression. Eradication is occasionally attempted when a foreign pest has been accidentally introduced but is not yet established in an area. Such eradication strategies often are supported by the Government. Mediterranean fruit fly, gypsy moth, and fire ant control programs are examples. In indoor areas, eradication is a more common goal. Enclosed environments usually are smaller, less complex, and more easily controlled than outdoor areas. In many enclosed areas, such as dwellings; schools; office buildings; and health care, food processing, and food preparation facilities, certain pests cannot or will not be tolerated.

Threshold Levels

Thresholds are the levels of pest populations at which you should take pest control action if you want to prevent the pests in an area from causing unacceptable injury or harm. Thresholds may be based on esthetic, health, or economic considerations. These levels, which are known as "action thresholds," have been determined for many pests. A threshold often is set at the level where the economic losses caused by pest damage, if the pest population continued to grow, would be greater than the cost of controlling the pests. These types of action thresholds sometimes are called "economic thresholds."

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In some pest control situations, the threshold level is zero: even a single pest in such a situation is unreasonably harmful. For example, the presence of any rodents in food processing facilities forces action. In homes, people generally take action to control some pests, such as rodents or roaches, even if only one or a few have been seen.

Pest Monitoring

In most pest control situations, the area to be protected should be monitored (checked or scouted) often. Regular monitoring can answer several important questions:

- What kinds of pests are present?
- Are the numbers great enough to warrant control?
- When is the right time to begin control?
- Have the control efforts successfully reduced the number of pests?

Monitoring of insect, insect-like, mollusk, and vertebrate pests usually is done by trapping or by scouting. Monitoring of weed pests usually is done by visual inspection. Monitoring for microbial pests is done by looking for the injury or damage they cause.

Monitoring also can include checking environmental conditions in the area that is being managed. Temperature and moisture levels, especially humidity, are often important clues in predicting when a pest outbreak will occur or will hit threshold levels.

Monitoring is not necessary in situations where a pest is continually present and the threshold is zero. For example, there is zero tolerance for the presence of bacteria in operating rooms and other sterile areas of health care facilities. In these situations, routine pest control measures are taken to prevent pests from entering an area and to eradicate any pests that may be present.

Avoiding Harmful Effects

Pest control involves more than simply identifying a pest and using a control tactic. The treatment site, whether it is an outdoor area or inside a structure, usually contains other living organisms (such as people, animals, and plants) and nonliving surroundings (such as air, water, structures, objects, and surfaces). All of these could be affected by the pest control measures you choose. Unless you consider the possible effects on the entire system within which the pest exists, your pest control effort could

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cause harm or lead to continued or new pest problems. Rely on your own good judgment and, when pesticides are part of the strategy, on the pesticide labeling.

Most treatment sites are disrupted to some degree by pest control strategies. The actions of every type of organism or component sharing the site usually affect the actions and well-being of many others. When the balance is disrupted, certain organisms may be destroyed or reduced in number, and others — sometimes the pests — may dominate.

Integrated Pest Management

Integrated pest management is the combining of appropriate pest control tactics into a single plan (strategy) to reduce pests and their damage to an acceptable level. Using many different tactics to control a pest problem tends to cause the least disruption to the living organisms and nonliving surroundings at the treatment site. Relying only on pesticides for pest control can cause pests to develop resistance to pesticides, can cause outbreaks of other pests, and can harm surfaces or non-target organisms. With some types of pests, use of pesticides as the only tactic will achieve very poor control. To solve pest problems, you must:

- identify the pest or pests and determine whether control is warranted for each.

- determine your pest control goal(s).
- know what control tactics are available.
- evaluate the benefits and risks of each tactic or combination of tactics.
- choose a strategy that will be most effective and will cause the least harm to people and the environment.
- use each tactic in the strategy correctly.
- observe local, State, and Federal regulations that apply to the situation.

The strategy you choose will depend on the pest you have identified and the kind and amount of control you need.

1. Natural Controls: Some natural forces act on all organisms, causing the populations to rise and fall. These natural forces act independently of humans and may either help or hinder pest control. You may not be able to alter the action of natural forces on a pest population, but you should be aware of their

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influence and take advantage of them whenever possible. Natural forces that affect pest populations include climate, natural enemies, natural barriers, availability of shelter, and food and water supplies.

2. Climate: Weather conditions, especially temperature, day length, and humidity, affect pests' activity and their rate of reproduction. Pests may be killed or suppressed by rain, freezing temperatures, drought, or other adverse weather. Climate also affects pests indirectly by influencing the growth and development of their hosts. A population of plant-eating pests is related to growth of its host plants. Unusual weather conditions can change normal patterns so that increased or decreased damage results.

3. Natural enemies: Birds, reptiles, amphibians, fish, and mammals feed on some pests and help control their numbers. Many predatory and parasitic insect and insect-like species feed on other organisms, some of which are pests. Pathogens often suppress pest populations.

4. Geographic barriers: Features such as mountains and large bodies of water restrict the spread of many pests. Other features of the landscape can have similar effects.

5. Food and water supply: Pest populations can thrive only as long as their food and water supply lasts. Once the food source - plant or animal - is exhausted, the pests die or become inactive. The life cycle of many pests depends on the availability of water.

6. Shelter: The availability of shelter can affect some pest populations. Overwintering sites and places to hide from predators are important to the survival of some pests.

7. Applied Controls: Unfortunately, natural controls often do not control pests quickly or completely enough to prevent unacceptable injury or damage. Then other control measures must be used. Those available include:

- host resistance,
- biological control,
- cultural control,
- mechanical control,

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- sanitation, and
- chemical control.

a. Host resistance: Some plants, animals, and structures resist pests better than others. Some varieties of plants, wood, and animals are resistant to certain pests. Use of resistant types, when available, helps keep pest populations below harmful levels by making conditions less favorable for the pests. Host resistance works in three main ways:

- Chemicals in the host repel the pest or prevent the pest from completing its life cycle.
- The host is more vigorous or tolerant than other varieties and thus less likely to be seriously damaged by pest attacks.
- The host has physical characteristics that make it more difficult to attack.

b. Biological control: Biological control involves the use of natural enemies — parasites, predators, and pathogens. You can supplement this natural control by releasing more of a pest's enemies into the target area or by introducing new enemies that were not in the area before. Biological control usually is not eradication. The degree of control fluctuates. There is a time lag between pest population increase and the corresponding increase in natural controls. But, under proper conditions, sufficient control can be achieved to eliminate the threat to the plant or animal to be protected. Biological control also includes methods by which the pest is biologically altered, as in the production and release of large numbers of sterile males and the use of pheromones or juvenile hormones.

Pheromones can be useful in monitoring pest populations. Placed in a trap, for example, they can attract the insects in a sample area so that pest numbers can be estimated. Pheromones also can be a control tool.

Sometimes a manufactured copy of the pheromone that a female insect uses to attract males can be used to confuse males and prevent mating, resulting in lower numbers of pests. Applying juvenile hormones to an area

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can reduce pest numbers by keeping some immature pests from becoming normal, reproducing adults.

c. Cultural control: Cultural practices sometimes are used to reduce the numbers of pests that are attacking cultivated plants. These practices alter

the environment, the condition of the host plant, or the behavior of the pest to prevent or suppress an infestation. They disrupt the normal relationship between the pest and the host plant and make the pest less likely to survive, grow, or reproduce. Common cultural practices include rotating crops, cultivating the soil, varying time or planting or harvesting, planting trap crops, adjusting row width, and pruning, thinning, and fertilizing cultivated plants.

d. Mechanical (physical) control: Devices, machines, and other methods used to control pests or alter their environment are called mechanical or physical controls. Traps, screens, barriers, fences, nets, radiation, and electricity sometimes can be used to prevent the spread of pests into an area. Lights, heat, and refrigeration can alter the environment enough to suppress or eradicate some pest populations. Altering the amount of water, including humidity, can control some pests, especially insects and disease agents.

e. Sanitation: Sanitation practices help to prevent and suppress some pests by removing the pests themselves or their sources of food and shelter. Urban and industrial pests can be reduced by improving cleanliness, eliminating pest harborage, and increasing the frequency of garbage pickup. Management of pests attacking domestic animals is enhanced by good manure management practices. Carryover of agricultural pests from one planting to the next can be reduced by removing crop residues. Other forms of sanitation that help prevent pest spread include using pest-free seeds or transplants and decontaminating equipment, animals, and other possible carriers before allowing them to enter a pest-free area or leave an infested area. The proper design of food-handling areas can reduce access and shelter for many pests.

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f. Chemical control: Pesticides are chemicals used to destroy pests, control their activity, or prevent them from causing damage. Some pesticides either attract or repel pests. Chemicals that regulate plant growth or remove foliage also are classified as pesticides. Pesticides are generally the fastest way to control pests. In many instances, they are the only tactic available

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3.0 FUNDAMENTALS OF PESTICIDES APPLICATION

Pesticides are a general category of agricultural chemicals that kill or otherwise control insects, rodents, nematodes, and weeds, and that prevent or control the infection of plants by viruses, fungi, and bacteria. Use of pesticides is not a new concept. Pesticides have been used for centuries to control pests that destroy crops. Today, we use dry, soluble, and liquid pesticides in agriculture to protect food and fiber,

and domestically to protect humans, pets, and property from mosquitoes, ants, termites, and other pests. The pros and cons of using pesticides are the subject of much debate among agricultural producers, chemical companies, government regulators, and environmental activist groups. One of the primary concerns is the long-term effect of synthetic pesticides on the environment, ecosystem, and human health. However, the benefit of pesticide use in recent history is evident, especially in terms of food production and cost.

According to the U.S. Department of Agriculture, when compared to other countries, Americans spend the lowest proportion of their family income— 10.1 percent— on food. In contrast, families in France, Italy, and the Philippines spend approximately 14.6, 15.2, and 51.3 percent, respectively, of their income on food. The U.S. government regulates the use of pesticides and the manner that they are stored, handled, and applied. The primary law governing pesticide use is the Federal Insecticide, Fungicide and Rodenticide Act of 1947. It has been amended several times, most recently by the Food Quality Protection Act of 1996. The primary intent of this law is to provide federal control of pesticide distribution, sale, and use. Added amendments require that pesticide applicators be certified and that all pesticides used in the United States be registered by the Environmental Protection Agency. Registration assures that pesticides will be properly labeled and that, if used in accordance with label specification, will not cause unreasonable harm to the environment.

3.1 Pesticide storage and handling

When storing, transporting, mixing, loading, or applying pesticides, or cleaning pesticide spills, it is good practice to treat all pesticides as though they are toxic. Read 19

all pesticide labels before use, and train all employees or pesticide handlers on personal protection procedures. Always keep unauthorized people, especially children, away from pesticide mixing, handling, and storage areas. Following are additional suggestions to use while storing and handling pesticides.

3.2 Storage of pesticide

Proper storage of pesticides can greatly reduce the risk of unauthorized personnel, especially children, from contacting, spilling, or ingesting pesticide material.

1. Keep the storage area locked. Pesticides can be very harmful when in the wrong hands.
2. Post storage areas and buildings with signs reading "Danger - Pesticides." The signs will also inform fire fighters that pesticides are present.
3. Always keep children, animals, and unauthorized persons away from pesticides.
4. Store pesticides in well ventilated, dry areas.
5. Don't keep large amounts of pesticides on hand; only purchase the amount you need.
6. Keep an inventory of pesticides and other chemicals, and their respective

locations.

7. Keep pesticides in their original containers. Never put them in unmarked or food containers.
8. Never store pesticides with food products, livestock feed, or fertilizer.
9. Store personal protective equipment in a clean area away from pesticides.
10. Periodically check pesticide containers for leaks or corrosion; some pesticides are caustic.

3.3 Loading and transportation of pesticide

Use caution when loading and transporting pesticides. Make sure handlers know how to properly load and secure pesticide containers, and know how to react to pesticide accidents.

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1. Inspect the vehicle being used to transport the pesticides. Make sure it is functioning properly.
2. Transport the pesticides in the back of the truck bed or in locations away from passengers.
3. Secure pesticide containers to ensure that they will not roll around or fall out. Prevent the containers from moving by tying down, blocking, and bracing them.
4. During loading, check the containers for leaks, make sure caps are secure, read the labels, and inventory the number and type of containers being transported.
5. Never transport pesticides with food or feed.
6. Never allow anyone to ride with the pesticides.
7. Never carry pesticides in the passenger seating area.
8. Be prepared for a spill during transportation.
9. Carry a safety kit for use during clean up. The kit should contain an index card with emergency numbers, duct tape, shovel, respirator, goggles, rubber gloves, protective clothing, soap, and wooden dowels to plug leaks. Also carry kitty litter or sand as an absorbent material.
10. If a spill happens, control and contain it. Put on safety equipment, and dike off the area. Contact the proper authorities for help.

3.4 Mixing of pesticide for use

Only authorized and trained personnel should be allowed to mix pesticides. Treat all pesticides as if they are potentially dangerous.

1. Mix pesticides carefully and accurately, using only the recommended amount specified on the label.
2. Read the label carefully, and follow the directions exactly.
3. Utilize appropriate personal protective equipment including gloves, splash-proof goggles or face shield, and protective clothing.
4. Keep hands away from the face, head, and neck when mixing.
5. Open liquids on a level surface and below eye-level to avoid spilling and

splashing.

6. Pour liquids below eye-level and as close to the ground as possible.

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7. Do not try to pour from a container that is too heavy.

8. Open pesticide powders with scissors to avoid dusts.

9. Use proper measuring tools when mixing pesticides.

10. Mix pesticides outside or in a well lit and ventilated area.

3.5 Loading and mixing into large tanks

Most loading and mixing of pesticides into large tanks such as agricultural machinery sprayers requires special caution due to the effect of wind drift and the potential for contamination into clean water supplies.

1. Recognize weather conditions when loading and mixing pesticides, especially the direction the wind is blowing.

2. Stand with your back to the wind so the pesticide will be blown away from you, not on you.

3. Keep your head well above the tank opening to prevent pesticides from splashing in your face.

4. Do not use your hands to stir pesticides or retrieve something that has fallen into the tank.

5. Close all containers as you finish with them.

6. Select the right equipment; use and maintain equipment properly.

7. Install an approved anti-siphoning device to prevent back siphoning into the water supply.

8. Maintain an air gap between the source and the spray tank.

9. Fill the spray tank with water and add the pesticide last, preventing the fill hose from becoming contaminated.

10. Add pesticides to the water-filled tank away from the water source. If possible, add pesticides to tanks while the sprayer is in the field.

11. If a pesticide must be added first, secure the hose to the top of the tank out of the liquid so it is not drawn in. Use check valves to prevent backflow into the water supply.

12. Avoid run-overs. Never leave filling operations unattended.

13. Protect well heads; never store chemicals near wells.

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3.6 Applying pesticide for use

It is unlawful to apply pesticides in a manner inconsistent with label instructions.

For agricultural producers, strict re-entry restrictions have been established for fields that have been recently sprayed with pesticides. Applicators should understand the risks and take proper measures to avoid them.

1. Set application equipment for the correct delivery rate, and operate equipment at

the recommended speed for proper coverage.

2. Check the sprayer for any loose connections or worn hoses.
3. Know and maintain the proper pressure and speed to avoid damage.
4. Check weather conditions when applying pesticides. It is against the law to apply pesticides on windy days when they might drift on to nearby fields.
5. Turn the sprayer off when turning around at the end of rows.
6. Turn the sprayer off when you are moving from field to field.
7. Don't apply pesticides when heavy rain is likely. Rain may wash pesticide residue into non-target areas.
8. Should the equipment become clogged or not work properly when spraying, take the necessary precautions when fixing it.
9. Wear appropriate personal protective equipment including gloves, eye protection, respirators, and special clothing.
10. Use a brush or soft copper wire to clean out clogged nozzles. Never use your mouth.
11. After spraying fields, obey re-entry periods. Post warnings around treated areas, and verbally warn others to stay out until it is safe to re-enter.
12. Use extreme caution when applying chemicals near bodies of water. Use grass buffer zones to minimize pesticide run-off.

3.7 Cleaning of pesticide in the event of spillage

In the event of a pesticide spill, use proper cleaning practices to avoid contaminating workers, animals, equipment, tools, and other objects. Make sure the appropriate cleaning materials are available near storage and handling facilities.

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1. Pressure wash or triple rinse all empty containers and flush hoses.
2. Put the rinse water into a tank for use on a labeled crop.
3. Safely dispose of empty containers and excess chemicals as suggested by the manufacturer, dealer, or agricultural authority.
4. Know the laws for disposing of pesticide containers. Never dump them where they could pollute groundwater, wells, or streams, or could contact people and animals.
5. Clean up all spills and leaks immediately. Some pesticides are caustic and could cause damage to floors and other structures.
6. Keep others away from the spill area, and make sure it does not move off-site.
7. Keep clean-up supplies such as a containment drum, kitty litter, sand, sawdust, shovel, broom, and dustpan in your storage facility and ready to use.
8. Clean up spills with soil, sand, rags, or paper towels.
9. Scoop up dry, contaminated material, place it into a leak-proof container, and properly dispose of it.
10. Rinse the area, but do not let the rinse water flow into ponds or streams.
11. Clean all equipment and protective clothing when finished.

12. Wash hands and face before eating, drinking, smoking, or chewing gum or tobacco.

13. Make sure you wash your hands before using the toilet at work.

3.8 How to handle pesticide poisoning

Despite all efforts to store and handle pesticides safely, accidents can and do happen. Knowing how to react to pesticide poisoning is essential. Be prepared for pesticide accidents. Have a written plan-of-action to respond quickly. The life you save may be your own.

1. Know the signs and symptoms of pesticide poisoning for all materials you use.
2. Make sure your employees and family members know how to recognize signs of overexposure.

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3. Post the telephone number of the Poison Control Center, your physician, and the nearest hospital in close proximity to pesticide handling and storage areas. When you call, have the label handy.

4. If you go to the physician or hospital for a poisoning emergency, take an original container or label with you.

5. Check the pesticide's Materials Safety Data Sheet for more detail on first aid procedures.

3.9 Disposal of pesticides wastes

After pesticide application, make sure that the treated area is clean and does not contain any pesticide waste, container or envelope and equipment is empty and clean. Never reuse empty envelopes, pesticide packing and containers for drinking and eating purposes for animals or people. All empty containers should be disposed of with proper care. The recommendations for this purpose are as follows:

1. Wash and puncture metal cans and drums and bury them.
2. Wash and puncture plastic containers, burn them and bury.
3. Cardboard packaging should be burnt or dumped carefully.
4. Plastic bags should be burnt in the areas away from dwellings, crops and stores etc.
5. Use mask during burning and keep children away from its smoke.

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4.0 Environmental Impact of pesticides

The impact of pesticides consists of the effects of pesticides on non-target species. Pesticides are chemical preparations used to kill fungal or animal pests. Over 98% of sprayed insecticides and 95% of herbicides reach a destination other than their target species, because they are sprayed or spread across entire agricultural fields. Runoff can carry pesticides into aquatic environments while wind can carry them to other fields, grazing areas, human settlements and undeveloped areas, potentially

affecting other species. Other problems emerge from poor production, transport and storage practices. Over time, repeated application increases pest resistance, while its effects on other species can facilitate the pest's resurgence.

Each pesticide or pesticide class comes with a specific set of environmental concerns. Such undesirable effects have led many pesticides to be banned, while regulations have limited and/or reduced the use of others. The global spread of pesticide use, including the use of older/obsolete pesticides that have been banned in some jurisdictions, has increased overall.

4.1 Effect of pesticides in Air

Pesticides can contribute to air pollution. Pesticide drift occurs when pesticides suspended in the air as particles are carried by wind to other areas, potentially contaminating them. Pesticides that are applied to crops can volatilize and may be blown by winds into nearby areas, potentially posing a threat to wildlife. Weather conditions at the time of application as well as temperature and relative humidity change the spread of the pesticide in the air. As wind velocity increases so does the spray drift and exposure. Low relative humidity and high temperature result in more spray evaporating. The amount of inhalable pesticides in the outdoor environment is therefore often dependent on the season. Also, droplets of sprayed pesticides or particles from pesticides applied as dusts may travel on the wind to other areas, or pesticides may adhere to particles that blow in the wind, such as dust particles. Ground spraying produces less pesticide drift than aerial spraying does. Farmers can employ a buffer zone around their crop, consisting of empty land or non-crop plants such as evergreen trees to serve as windbreaks and absorb the pesticides, preventing drift into other

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areas. Pesticides that are sprayed on to fields and used to fumigate soil can give off chemicals called volatile organic compounds, which can react with other chemicals and form a pollutant called tropospheric ozone. Pesticide use accounts for about 6 percent of total tropospheric ozone levels.

4.2 Effect of pesticides on water

Pesticide residues have also been found in rain and groundwater. Pesticide impacts on aquatic systems are often studied using a hydrology transport model to study movement and fate of chemicals in rivers and streams. As early as the 1970s quantitative analysis of pesticide runoff was conducted in order to predict amounts of pesticide that would reach surface waters. There are four major routes through which pesticides reach the water: it may drift outside of the intended area when it is sprayed, it may percolate, or leach through the soil, it may be carried to the water as runoff, or it may be spilled, for example accidentally or through neglect. They may also be carried to water by eroding soil. Factors that affect a pesticide's ability to contaminate water include its water solubility, the distance from an application site to a body of water, weather, soil type, presence of a growing crop, and the method used to apply the

chemical.

Fig 1: Pesticide pathways

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4.3 Effect of pesticides in the soil

The extensive use of pesticides in agricultural production can degrade and damage the community of microorganisms living in the soil, particularly when these chemicals are overused or misused as chemical compounds build up in the soil. The full impact of pesticides on soil microorganisms is still not entirely understood; many studies have found deleterious effects of pesticides on soil microorganisms and biochemical processes, while others have found that the residue of some pesticides can be degraded and assimilated by microorganisms. The effect of pesticides on soil microorganisms is impacted by the persistence, concentration, and toxicity of the applied pesticide, in addition to various environmental factors. This complex interaction of factors makes it difficult to draw definitive conclusions about the interaction of pesticides with the soil ecosystem. In general, long-term pesticide application can disturb the biochemical processes of nutrient cycling. Many of the chemicals used in pesticides are persistent soil contaminants, whose impact may endure for decades and adversely affect soil conservation.

The use of pesticides decreases the general biodiversity in the soil. Not using the chemicals results in higher soil quality, with the additional effect that more organic matter in the soil allows for higher water retention. This helps increase yields for farms in drought years, when organic farms have had yields 20-40% higher than their conventional counterparts. A smaller content of organic matter in the soil increases the amount of pesticide that will leave the area of application, because organic matter binds to and helps break down pesticides.

Degradation and sorption are both factors which influence the persistence of pesticides in soil. Depending on the chemical nature of the pesticide, such processes control directly the transportation from soil to water, and in turn to air and our food. Breaking down organic substances, degradation, involves interactions among microorganisms in the soil. Sorption affects bioaccumulation of pesticides which are dependent on organic matter in the soil. Weak organic acids have been shown to be weakly sorbed by soil, because of pH and mostly acidic structure. Sorbed chemicals have been shown to be less accessible to microorganisms. Aging mechanisms are

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poorly understood but as residence times in soil increase, pesticide residues become more resistant to degradation and extraction as they lose biological activity.

4.4 Effects of pesticides on plants

Nitrogen fixation, which is required for the growth of higher plants, is hindered by pesticides in soil. The insecticides DDT, methyl parathion, and

especially pentachlorophenol have been shown to interfere with legume-rhizobium chemical signaling. Reduction of this symbiotic chemical signaling results in reduced nitrogen fixation and thus reduced crop yields. Root nodule formation in these plants saves the world economy \$10 billion in synthetic nitrogen fertilizer every year.

Pesticides can kill bees and are strongly implicated in pollinator decline, the loss of species that pollinate plants, including through the mechanism of Colony Collapse Disorder, in which worker bees from a beehive or western honey bee colony abruptly disappear. Application of pesticides to crops that are in bloom can kill honeybees, which act as pollinators. On the other side, pesticides have some direct harmful effect on plant including poor root hair development, shoot yellowing and reduced plant growth.

4.5 Effect on Animals

Many kinds of animals are harmed by pesticides, leading many countries to regulate pesticide usage through Biodiversity Action Plans. Animals including humans may be poisoned by pesticide residues that remain on food, for example when wild animals enter sprayed fields or nearby areas shortly after spraying.

Pesticides can eliminate some animals' essential food sources, causing the animals to relocate, change their diet or starve. Residues can travel up the food chain; for example, birds can be harmed when they eat insects and worms that have consumed pesticides. Earthworms digest organic matter and increase nutrient content in the top layer of soil. They protect human health by ingesting decomposing litter and serving as bioindicators of soil activity. Pesticides have had harmful effects on growth and reproduction on earthworms. Some pesticides can bioaccumulate, or build up to toxic levels in the bodies of organisms that consume them over time, a phenomenon that impacts species high on the food chain especially hard.

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a. Birds: Reductions in bird populations have been found to be associated with times and areas in which pesticides are used. Some pesticides come in granular form. Wildlife may eat the granules, mistaking them for grains of food. A few granules of a pesticide may be enough to kill a small bird. Herbicides may endanger bird populations by reducing their habitat. DDE-induced egg shell thinning has especially affected European and North American bird populations. From 1990 to 2014 the number of common farmland birds has declined in the European Union as a whole and in France, Belgium and Sweden; in Germany, which relies more on organic farming and less on pesticides the decline has been slower; in Switzerland, which does not rely much on intensive agriculture, after a decline in the early 2000s the level has returned to the one of 1990. In another example, some types of fungicides used in peanut farming are only slightly toxic to birds and mammals, but may kill earthworms, which can in turn reduce populations of the birds and mammals that feed on them.

b. Aquatic life: Fish and other aquatic biota may be harmed by pesticide-

contaminated water. Pesticide surface runoff into rivers and streams can be highly lethal to aquatic life, sometimes killing all the fish in a particular stream. Application of herbicides to bodies of water can cause fish kills when the dead plants decay and consume the water's oxygen, suffocating the fish. Herbicides such as copper sulfate that are applied to water to kill plants are toxic to fish and other water animals at concentrations similar to those used to kill the plants. Repeated exposure to sublethal doses of some pesticides can cause physiological and behavioral changes that reduce fish populations, such as abandonment of nests and broods, decreased immunity to disease and decreased predator avoidance.

Application of herbicides to bodies of water can kill plants on which fish depend for their habitat. Pesticides can accumulate in bodies of water to levels that kill off zooplankton, the main source of food for young fish. Pesticides can also kill off insects on which some fish feed, causing the fish to travel farther in search of food and exposing them to greater risk from predators. The faster a given

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pesticide breaks down in the environment, the less threat it poses to aquatic life. Insecticides are typically more toxic to aquatic life than herbicides and fungicides.

c. Amphibians: In the past several decades, amphibian populations have declined across the world, for unexplained reasons which are thought to be varied but of which pesticides may be a part. Pesticide mixtures appear to have a cumulative toxic effect on frogs. Tadpoles from ponds containing multiple pesticides take longer to metamorphose and are smaller when they do, decreasing their ability to catch prey and avoid predators. Exposing tadpoles to the organochloride endosulfan at levels likely to be found in habitats near fields sprayed with the chemical kills the tadpoles and causes behavioral and growth abnormalities. The herbicide atrazine can turn male frogs into hermaphrodites, decreasing their ability to reproduce. Both reproductive and non-reproductive effects in aquatic reptiles and amphibians have been reported. Crocodiles, many turtle species and some lizards lack sex-distinct chromosomes until after fertilization during organogenesis, depending on temperature. Embryonic exposure in turtles to various PCBs causes a sex reversal. Across the United States and Canada disorders such as decreased hatching success, feminization, skin lesions, and other developmental abnormalities have been reported.

d. Humans: Pesticides can enter the body through inhalation of aerosols, dust and vapor that contain pesticides; through oral exposure by consuming food/water; and through skin exposure by direct contact. Pesticides secrete into soils and groundwater which can end up in drinking water, and pesticide spray can drift and pollute the air. The effects of pesticides on human health depend on the toxicity of the chemical and the length and magnitude of exposure. Farm workers and their families experience the greatest exposure to agricultural

pesticides through direct contact. Every human contains pesticides in their fat cells.

Children are more susceptible and sensitive to pesticides, because they are still developing and have a weaker immune system than adults. Children may be more exposed due to their closer proximity to the ground and tendency to put

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unfamiliar objects in their mouth. Hand to mouth contact depends on the child's age, much like lead exposure. Children under the age of six months are more apt to experience exposure from breast milk and inhalation of small particles.

Pesticides tracked into the home from family members increase the risk of exposure. Toxic residue in food may contribute to a child's exposure. The chemicals can bioaccumulate in the body over time.

Exposure effects can range from mild skin irritation to birth defects, tumors, genetic changes, blood and nerve disorders, endocrine disruption, coma or death. Developmental effects have been associated with pesticides. Recent increases in childhood cancers in throughout North America, such as leukemia, may be a result of somatic cell mutations. Insecticides targeted to disrupt insects can have harmful effects on mammalian nervous systems. Both chronic and acute alterations have been observed in exposes. DDT and its breakdown product DDE disturb estrogenic activity and possibly lead to breast cancer. Fetal DDT exposure reduces male penis size in animals and can produce undescended testicles. Pesticide can affect fetuses in early stages of development, in utero and even if a parent was exposed before conception. Reproductive disruption has the potential to occur by chemical reactivity and through structural changes.

4.6 Persistent organic pollution

Persistent organic pollutants (POPs) are compounds that resist degradation and thus remain in the environment for years. Some pesticides, including aldrin, chlordane, DDT, dieldrin, endrin, heptachlor, hexachlorobenzene, mirex and toxaphene, are considered POPs. Some POPs have the ability to volatilize and travel great distances through the atmosphere to become deposited in remote regions. Such chemicals may have the ability to bioaccumulate and biomagnify and can biomagnify (i.e. become more concentrated) up to 70,000 times their original concentrations. POPs can affect non-target organisms in the environment and increase risk to humans by disruption in the endocrine, reproductive, and respiratory systems.

4.7 Pest resistance

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Pests may evolve to become resistant to pesticides. Many pests will initially be very susceptible to pesticides, but following mutations in their genetic makeup become resistant and survive to reproduce. Resistance is commonly managed through pesticide

rotation, which involves alternating among pesticide classes with different modes of action to delay the onset of or mitigate existing pest resistance.

4.8 Pest rebound and secondary pest outbreak

Non-target organisms can also be impacted by pesticides. In some cases, a pest insect that is controlled by a beneficial predator or parasite can flourish should an insecticide application kill both pest and beneficial populations. A study comparing biological pest control and pyrethroid insecticide for diamondback moths, a major cabbage family insect pest, showed that the pest population rebounded due to loss of insect predators, whereas the biocontrol did not show the same effect. Likewise, pesticides sprayed to control mosquitoes may temporarily depress mosquito populations, they may result in a larger population in the long run by damaging natural controls. This phenomenon, wherein the population of a pest species rebounds to equal or greater numbers than it had before pesticide use, is called pest resurgence and can be linked to elimination of its predators and other natural enemies.

Loss of predator species can also lead to a related phenomenon called secondary pest outbreaks, an increase in problems from species that were not originally a problem due to loss of their predators or parasites. An estimated third of the 300 most damaging insects in the US were originally secondary pests and only became a major problem after the use of pesticides. In both pest resurgence and secondary outbreaks, their natural enemies were more susceptible to the pesticides than the pests themselves, in some cases causing the pest population to be higher than it was before the use of pesticide.

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PESTICIDES

Case studies of global abuse and misuse of pesticides

Pesticides can be defined as substances or mixture of pesticides that can be used to prevent, control, kill,

repel or mitigate the harmful effects of pests.

They could be in granular, liquid, spray or powdery form

What is a pest

Pest in this context include Fungi, Weeds, Bacteria, Rodents and any other organism which could cause

distortion in agriculture

Know Pesticide Abuse -

Know Pesticide Misuse

Pesticide Misuse - Can be defined as the use of pesticides in a manner that contradicts labeled instruction, local regulation or best management practices

Examples of Pesticide misuse

1. Possession of unregistered pesticides

2. Spraying of pesticides in windy conditions
3. Application of pesticides without wearing protection gear
4. Storing pesticides in on unmarked or unlabeled container
5. Indiscriminate disposal of used pesticides container
6. Applying pesticide without reading the instructions for its use

Pesticide Abuse- Can be defined as the intentional and excessive use of pesticides for purposes that are

not agricultural or Horticultural

°•Reasons for pesticide misuse and abuse

- Ignorance/Lack of information
- Misleading information on the pesticide
- Profit maximization
- Lack of check and balances: Regulatory bodies not doing their job
- Ignorance of the consequence
- Habits
- Lack of alternative
- Government Policies

CASE STUDIES

The Minamata Bay Disaster

This started in 1930 but got discovered in 1950 in Chisso which is located in Japan.

Nature of abuse: Methylmercury is registered into a bay

Short term Effects:

Loss of hearing /Eyesight

Numbness

Skin irritation

Weakness

Long term Effects:

Cancer

Birth defect

Psychological Trauma

Respiratory diseases

Agent Orange in Vietnam (1961-1971)

Agent Orange contained 2,4 Dichlorophenoxyacetic acid and 2,4,5 - T and was used by the American's

Ranch Hand Operation

It was called agent orange because of the container □ and have strippers .

Sparing agent

Short term consequences : skin irritation, skin diseases ,

Long term consequences : cancer

Effects : lost of soil degradation

Lost of biodiversity

Water contamination

Bio ammunition

Green revolution in punja (1960)

The era had an interest in family and wanted to grow crops to feed them , and these crops were rice and

wheat . The abuse came when some wanted to get this plants reduce for some self agriculture crops

growth

So they came with a Pesticide called DDT Dichloro diphenyl trichlorodethane Benzene hexachloride ,

This is the green revolution in Punija .